Question Paper **Total Mark: 20**

***Read the following carefully*:**

- This is a 24-hour take-home quiz.
- Answer all questions. **Please show your working. If you do not show any working, no marks will be awarded even if your answer is correct.**
- Hand-write your answers. Writing with gadgets such as Apple Pencil is allowed. If you are taking photos, ensure that the photos are clear and readable.
- Save your answers as a **PDF file**, and **name it using your student ID**, for example **EEE1234567.pdf**.
- Please download the cover page and answer script. Fill in the cover page. This question paper is not required to attached to the answer script for submission. Submit only the cover page with solutions.

Question 1 [6 marks]

Suppose that two balls are randomly selected from an urn containing 2 red, 3 white and 4 blue balls. Let

X = number of white balls,

Y = number of blue balls.

(i) Find the joint probability mass function of X and Y, $p(x, y)$, as well as the marginal probability mass function, $p_X(x)$ and $p_Y(y)$.

(ii) Find the conditional probability mass function of X given $Y = 0$.

Question 2 [4 marks]

The joint probability density function of X and Y are given by

$$f(x, y) = \begin{cases} 4e^{-x}e^{-4y}, & x > 0, y > 0 \\ 0, & \text{otherwise} \end{cases}.$$

Find $P(X < 2, Y < 2)$.

Question 3 [10 marks]

Let X_1, X_2 and X_3 be independent random variables representing lifetimes (in hours) of three key components of a device connected in parallel. X_i has an exponential distribution with mean 800 for i = 1, 2, 3. The device fails when and only when all three components fail. Let $Y_1 = \min(X_1, X_2, X_3)$ and $Y_2 = \max(X_1, X_2, X_3)$, so that the space of Y_1, Y_2 is $0 < y_1 < y_2 < \infty$.

(i) Find the marginal cumulative distribution functions, $F_{Y_1}(z)$ and $F_{Y_2}(z)$.

(ii) Compute the probability that the device fails after 1,600 hours.